

Quantum-Sound Factorization: On the Distribution and Structure of Irreducible Sets[†]

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Introduction - Why do we care?

- Most of modern digital security relies on the difficulty of *Integer Factorization*. The complexity of this problem lies in the randomness in the primes, the atoms of the integer universe.
- In 1994, the mathematician Peter Shor showed that a quantum computer can solve the integer factorization problem strikingly fast, making quantum computers a major threat to security.
- Ever since, mathematicians and cryptographers have been racing to find quantum-secure digital systems. Billions of dollars have been poured into the so-called field of post-quantum cryptography.
- In particular, it is important to discover and study mathematical structures that possess complex factorization properties, as they may be the basis of a quantum-secure system.

Introduction - History

- Mathematicians in the late 19th century began to understand the complexity of *Set Factorization*.
- A collection of sets can be equipped with a binary operation called the *Minkowski Sum* or *sumset*. The study of the factorization of sets under this operation is central to *Additive Combinatorics*, one of the most active fields of mathematics in the 21st century.
- In the 1960s, Hans-Heinrich Ostmann pioneered the study of *irreducible sets* – the analogue of prime numbers for sets.
- In 2005, Kim and Roush [3] showed that set factorization (and even testing a set for irreducibility) is NP-hard. The problem is likely also difficult for quantum computers.
- In [2], the authors show that one can extract randomness from sets that are not irreducible.
- **We lack the mathematical understanding of set factorization to create a cryptosystem based on it.** In this project, significant progress is made towards the required understanding.

In this project, the following progress is made:

- A generalized “prime number theorem” for sets is developed, answering an open question of Geroldinger and Tringali, and **establishing a complete understanding of the distribution of irreducible sets**.
- A novel irreducibility criterion is developed and analyzed. This criterion possesses the property that between 6 and 18% of all irreducible sets satisfy this criterion, making this **the strongest irreducibility criterion in any domain**. This criterion is **extremely useful to cryptographers**, as it poses a potential vulnerability in cryptosystems involving set factorization.
- Additional results on the structure and distribution of irreducible sets are developed.

Definitions

Let A and B be subsets of $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. The *sumset*, or *Minkowski sum*, of A and B is

$$A + B = \{a + b : a \in A, b \in B\}.$$

A set A is called *irreducible* if there are no sets B, C with $\min(|B|, |C|) > 1$ such that $A = B + C$. It turns out that we can restrict our study to sets containing 0 and still recover all information about sets not containing 0. We also restrict ourselves to finite sets, as for practical applications infinite sets are less useful. The collection of these sets under the sumset operation forms the restricted finitary power monoid $\mathcal{P}_{fin,0}(\mathbb{N}_0)$ of $\mathbb{N}_0$. The irreducible sets are the *atoms* of this monoid.

We are interested in counts of irreducible sets, so we let α_n denote the number of irreducible subsets of $[n] = \{0, 1, 2, \dots, n\}$, and $\alpha_{n,k}$ the number of such irreducible subsets of size k .

A Generalized Prime Number Theorem

- Shitov [4] originally showed that “almost all sets are irreducible”, i.e. $\alpha_n = 2^n(1 - o(1))$, where $o(1)$ represents a quantity tending to 0 as $n \rightarrow \infty$. This established the first prime number theorem for sets.
- Geroldinger and Bienvenu later improved this bound, proving in [1] that this quantity tends to 0 exponentially fast. In particular, there is a constant $c < 2$ such that, in the limit, $(c + o(1))^n$ subsets of $[n]$ are reducible. It is unknown what c is, apart from the bounds $1.7 < c < 2$.
- However, it was previously unknown as to how the size of a set affects its likelihood to be irreducible. The theorems on the next slide (stated loosely) resolve this question.

A Generalized Prime Number Theorem

Theorem (A.-Gotti-Lu, 2024)

Let $k(n)$ be such that $\frac{k(n)}{n} \rightarrow \delta$ as $n \rightarrow \infty$, where $0 < \delta < 1$. Then $\alpha_{n,k(n)} = \binom{n}{k(n)-1}(1 - o(1))$. In other words, almost all sets that are not too sparse nor too dense are irreducible.

Theorem (A., 2025)

Let $k(n)$ be such that $k(n)/n \rightarrow 0$ as $n \rightarrow \infty$. Then $\alpha_{n,k(n)} = \binom{n}{k(n)-1}(1 - o(1))$. In other words, almost all sparse sets are irreducible.

Theorem (A., 2025)

Let $k(n) = n - o(\sqrt{n})$. Then $\alpha_{n,k(n)} = o(1)\binom{n}{k(n)-1}$. In other words, almost all very dense sets are *reducible*.

A Novel Irreducibility Criterion

- The previous strongest polynomial-time checkable criterion for irreducibility is due to Ostmann, and it targets very few irreducible sets, in particular 0%.
- One big breakthrough in this project is the introduction and analysis of a stronger condition, targeting 6 – 18% of irreducible sets.
- The condition is polynomial-time checkable, naively being $O(n^2)$ for a subset of $[n]$ (this complexity can likely be improved with various optimizations).

Condition

A set S is irreducible if:

- $2m \notin S$, where $m = \min S_{>0}$.
- For all $s \in S_{>m}$, $s + S_{<s}$ is not a subset of S .

We call such a set *strongly irreducible*.

A Novel Irreducibility Criterion

The below theorem establishes the result mentioned on the previous slide.

Theorem (A., 2025)

Let s_n be the number of strongly irreducible subsets of $[n]$. Then

$$6.38\% \approx \frac{49}{768} \leq \lim_{n \rightarrow \infty} \frac{s_n}{2^n} \leq \frac{3}{16} = 18.75\%.$$

One can additionally prove that almost all sparse sets are strongly irreducible:

Theorem (A., 2025)

Let $s_{n,k}$ be the number of strongly irreducible subsets of $[n]$ of size k . If $k(n) \ll \frac{n}{\log n}$,

$$s_{n,k(n)} = \binom{n}{k(n)-1} (1 - o(1)).$$

Additional Results of Interest

A very interesting theoretical result is that irreducible sets approximately follow a binomial distribution. For each N , one can define a natural random variable X_N with probability mass function

$$\mathbb{P}(X_N = k) = \frac{\alpha_{n,k}}{\alpha_n}.$$

Then we have the following:

Theorem (A.-Gotti-Lu, 2024)

For each $N \in \mathbb{N}$, let $Y_N \sim \text{Bin}(N, 1/2)$. Then for every $r \in \mathbb{N}$ we have that the r th-moments of X_N and Y_N are roughly the same:

$$\lim_{N \rightarrow \infty} \frac{\mathbb{E}(X_N^r)}{\mathbb{E}(Y_N^r)} = 1.$$

We also prove an algebraic result that generalizes a theorem in [1], helping us understand the structure of monoids formed by sets under the sumset operation (details omitted for space reasons).

Concluding Remarks

- **This work is quite exciting, as it has many implications in both pure math and security in the quantum world.**
- It **significantly furthers our understanding** of irreducible sets, pushing closer to the implementation of a working cryptosystem using them.
- For example, it **identifies a major potential vulnerability** in such cryptosystems.
- It helps us understand the degree of randomness that can arise from irreducible sets, as well as further our knowledge of algebraic properties of numerical power monoids.
- The first paper for this project (joint with Gotti and Lu) is on arXiv (2412.05857) and submitted for publication. The second paper (individual) is soon to be submitted.

Selected References

- [1] Pierre-Yves Bienvenu and Alfred Geroldinger. “On algebraic properties of power monoids of numerical monoids”. In: *Israel Journal of Mathematics* (2022). URL: <https://api.semanticscholar.org/CorpusID:248496889>.
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- [3] Ki Hang Kim and Fred W. Roush. *Factorization of polynomials in one variable over the tropical semiring*. 2005. arXiv: math/0501167 [math.CO]. URL: <https://arxiv.org/abs/math/0501167>.
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